

Enhancing Brain Performance by Externalizing Thought

Ars longa vita brevis.

—ATTRIBUTED TO HIPPOCRATES

Science works for evil when its effect is to provide toys for the rich, and works for good when its effect is to provide necessities for the poor.

—FREEMAN DYSON¹

Technology is a mode of revealing. [It] makes the demand on us to think in another way.

—MARTIN HEIDEGGER²

In the textile industry of nineteenth-century Britain, the invention of the “knitting frame” and “shearing frame” made the cloth manufacturing process simpler so that workers with limited training could be employed. At the same time, these innovations threatened the weavers and artisans in Nottinghamshire, Yorkshire, and Lancashire who had spent years learning their craft. These highly skilled people saw the new machines as the cause of their decreasing income and lost jobs, so they picked up sledgehammers, invaded the mills during the night, and wrecked the machines. Ned Ludd, a young apprentice, was rumored to be the first attacker, and his followers called themselves Luddites. Between 1811 and 1816, Luddites raided and burned down dozens of mills and damaged machinery in hundreds of them. They also lobbied the

1. Dyson (1997).

2. Heidegger (1977).

British Parliament to restrict automated looms and knitting frames. However, the government sided with the mill owners, gunned down the rebels, and hanged the leaders or transported them to the colonies.³ The movement was crushed, but the word “Luddite” survived. Today, it is a blanket term used to describe technophobes who feel that industrialization, computerization, or new technologies in general do not always bring happiness. The Luddite movement can be viewed as a leaderless resistance to frightening technologies and consumerism that have damaged the livelihoods of part of the population.

Although the Luddites wrecked machines, likely their real goal was to dissuade the owners, the aggressive new class of manufacturers who drove the Industrial Revolution, from using cheap labor. The Luddites clearly knew who owned the technology that concerned them. Today, we are no longer sure who controls technology and who drives the need for constant innovation. An even more complex question is how much benefit we gain from it.⁴ Since President Franklin Roosevelt signed the Wages and Hours Bill in 1938, the five-day, forty-hour workweek has been the norm for many workers in the United States, followed by most other countries.⁵ Can we reduce our workload further because technological inventions allow it to happen and devote more time for recreation? Industries and offices can be shut off, and even season-dependent agricultural work can be operated by shifting groups of people. Therefore, in principle, we can determine the number of hours we work.

If bad decisions are human-made, it should be possible to unmake them. Several grassroots movements have tried to reduce the workload of humankind. Reasonable calculations show that current technological developments would allow people in industrialized nations to work only three or four days a week while maintaining the living standards of the 1960s. Such a reduction in

3. Sale (1995); Binfield (2004).

4. An irrational hostility to technology fueled the bombing acts of Theodore Kaczynski, the Harvard-educated mathematician, Assistant Professor at the University of California, Berkeley, and the infamous Unabomber. In his manifesto (*Industrial Society and Its Future*), published by the *New York Times* as part of his ransom demand, he said: “Industrial Revolution and its consequences have been a disaster for the human race.” [NeuroNote: A court-appointed psychiatrist declared that he suffered from paranoid schizophrenia, but Kaczynski dismissed this evaluation as “political diagnosis.”]

5. The seven-day grouping is an arbitrary epoch introduced by the Babylonians who believed there were seven planets in the solar system. To mark the end of the week, a worship of the gods was introduced. The Sabbath is on Friday, Saturday, or Sunday in Muslim, Jewish, and Christian religious traditions, respectively, and reflects the belief that God rests on the seventh day. The brain does not “feel” one week of time, and there is no clear indicator in the outside world to mark seven-day epochs. An exception may be in today’s urban areas, where air pollution varies significantly between work days and week days (Ward et al., 2015).

time spent working could address problems with unemployment, high carbon emissions, social inequalities, family care, and personal happiness for many. Yet, instead of reducing workloads, more of the world's population works long hours, and the full-time employment of women has increased over the past several decades.⁶

How did technology mysteriously descend on us? Who is to blame? It seems that nobody is in charge, and no one can stop it. Instead, all of us contribute to the chaotic system and impose more work on ourselves. We are driven by the short-term benefits of technology and economic envy of others instead of exploiting these innovations for our long-term well-being. When all of us are involved in a distributed process, nobody can make a decision on our behalf.⁷

In this chapter, I attempt to tackle this complex issue by examining how the externalization of brain function affects our brains and the brains of others. Some of the text may appear off-stream from the main topic of this book. However, without discussing these issues, it is not possible to understand why the relationship between episodic and semantic memories appear so different in humans than in other animals.⁸ I conclude that human-made instruments and other artifacts have become an extension of the action–perception loop, as well as the media through which abstract ideas emerge and spread quickly. Through this process, humans use their brains in a way that no other animal can approach. Without taking this externalization process into consideration, we cannot understand how complex thoughts and ideas are created by human

6. I am not competent to discuss the sociological, economic, or political implications of technology. Instead, I focus on the impact of externalized innovations on the brain in this chapter.

7. Naturally, humans complained about technological carnage after the two world wars. Similar hatred of technology was aroused in my generation, who witnessed the horror of napalm bombs in Vietnam. The hippie movement was largely fueled by the evil aspects of science. There is a catch-22 with the use of technology. A minority of people profit from it tremendously, while the majority benefits much less or not at all. Even if technology improves the quality of life for all, control of technical innovations comes with a disproportional benefit for those who are in control of the society. This economic scissor always makes the majority *feel* like exploited losers even when they benefit from it.

8. In Chapter 5, we discussed the idea that the evolutionary roots of episodic and semantic memory systems are the dead reckoning and landmark-based forms of navigation and that semantic knowledge requires multiple episodic events, through which process the spatiotemporal aspects of the events are stripped off (Buzsáki, 2005; Buzsáki and Moser, 2013). In contrast, Mishkin et al. (1998), studying the effects of early hippocampal injury in humans, conclude just the opposite. In their view, semantic information is primary and needed for constructing hippocampus-dependent episodes. The difference in emphasis may be the “shortcut” mechanisms in humans afforded by the ability to rapidly acquire semantic information through externalization of mental representation.

brains, which are not much more complicated structurally than the similarly large brains of some other species.

THOUGHTS ARE ACTIONS: EXTERNALIZATION OF MENTAL OPERATIONS

Brain outputs come in various flavors. The first one that comes to mind is muscular action to move the body and its sensors. A second output acts via the autonomic nervous system to affect our internal organs and glands. A third output is the release of hormones from the pituitary gland to affect growth, blood pressure, sexual hormones, thyroid hormones, metabolism, temperature, childbirth, lactation, and water/salt concentration in the kidneys. The fourth large category of brain output is thoughts, even if we do not intuitively think of thoughts as actions. A *thought*, which can be conceptualized as a buffer for a deferred action (Chapter 5), is useful only when it benefits the brain's owner.⁹

Imagination and thought rely largely on the hippocampus and prefrontal cortex. The prefrontal cortex can be considered an internalized offshoot of the motor cortex with a similar neural architecture. The motor and higher order prefrontal cortical areas (divided into medial, orbital, and insular areas) receive remarkably similar inputs and send similar outputs to their targets. The main difference is that while the primary motor cortex sends direct projections to the spinal cord to control the skeletal muscles, the prefrontal cortex instead targets autonomic and limbic sites, including the basal ganglia, basolateral amygdala, thalamus, hippocampus, and lateral hypothalamus.¹⁰ These projections can be viewed as corollary circuits through which prefrontal areas inform other higher order cortical areas and motivation or action-preparing structures about pending action plans, similar to the way that motor cortex informs sensory areas about ongoing actions. By analogy, we can conclude that prefrontal cortex projections compare pending or potential actions and their expected consequences against recalled information and desired goals. Instead of

9. Buddhist teaching emphasizes that physical and mental actions (thoughts) are related. "Karma" literally means action or doing. I thank Liset Menendez de la Prida for pointing out this relationship.

10. Other targets with notable projections from the medial prefrontal cortex in the rat are the mediolateral septum, dorsolateral periaqueductal gray, ventral tegmental area, parabrachial nucleus, nucleus tractus solitarius, rostral/caudal ventrolateral medulla, and even the thoracic spinal cord, with some notable differences across infralimbic (area 25), prelimbic (area 32), and dorsal anterior cingulate (area 24b) areas. Many of these connections are reciprocal (Gabbott et al., 2005).

directly evaluating actions, this hypothetical higher order corollary mechanism compares and evaluates the internal workings of the brain to form multiple-step action plans before any overt movement.

In humans and other great apes, several prefrontal areas contain large neurons in cortical layer 5; these are called *spindle cells*. These large cells are the first cousins of the similarly large layer-5 Betz cells of the primary motor cortex, which project to the spinal cord, another indication of the general architectural design of prefrontal cortical areas. These special-looking spindle cells were first observed in hominids, prompting several investigators to attribute to them unique functions in cognition and emotion control.¹¹ However, they have since been described in cetaceans, at higher densities than in humans, in elephants, and, to some extent, in rhesus monkeys and even raccoons. These neurons have the strongly myelinated, long axons needed for sending action potentials quickly across long distances in large animals. These anatomical considerations reveal that motor and prefrontal areas of the cortex share many anatomical features. The main functional difference is that while activity in the motor cortex leads to immediate action, activity in the prefrontal cortex may only simulate action, which we call *plans* and *imagination*.

Are Humans Superior?

Many investigators have sought to explain the “superior” qualities of *Homo sapiens* by some anatomical feature, such as brain size, neuronal numbers and densities, axonal connectivity, special cell types, or axon conduction speed. They found none. Our cousin, the Neanderthal, with whom we overlapped 50,000 years ago, had a bigger brain than modern humans do.¹² They surely had some special abilities compared to us, but what exactly they did better with so many neurons remains unexplained.

Modern humans who walked the planet 50,000–200,000 years ago had the same vocal cords, hands, and brains as we do today.¹³ Imagine that a child born 50,000 years ago could be swapped with a newborn of today. Would the child

11. Spindle cells are also known as *von Economo neurons*, named after Constantin von Economo (von Economo and Koskinas, 1929). Allman et al. (2002) studied these neurons extensively in several species.

12. Henneberg and Steyn (1993); Ruff et al. (1997); Bailey and Geary (2009). An excellent recent account of evolutionary comparison of the human brain with that of other animals is by Herculano-Houzel (2016).

13. The newly found skull at Jebel Irhoud, Morocco, was dated at 300,000 years old. The remains of the skulls of five individuals had modern facial traits, although the brain cases were elongated, with a somewhat smaller brain volume and not round like those in humans living today. Jebel Irhoud people are considered protomodern humans who controlled fire and hammered

born in New York City and teleported back in time to a loving family become a constructive member of a hunter-gatherer community? Conversely, would the teleported child from tens of thousands years ago, adopted by an average family in the United States, have the same chance of getting into college as my daughters? My answers are a resounding “yes.” I assume that our ancestors had the same cognitive and communication capacity as we do, and I would also venture that they had pretty much the same brain wiring and dynamics.¹⁴ If you side with me on this issue, then we also likely agree that something besides anatomical differences in the brain must explain why *Homo sapiens* managed to become the ruler of the world. In my view, that something else is that modern humans learned to externalize their minds by first making tools, and then inventing speech, writing, and art with ever-increasing sophistication. Along the way, they redesigned themselves into a cooperative, knowledge-sharing species.¹⁵

EXTERNALIZATION: A QUICK OVERVIEW

Homo habilis and *H. erectus* were already making bifacial hand axes to scrape meat from bones and butcher large animals 1.5 million years ago. Their technology remained relatively the same, with minor modifications, for a million years.¹⁶ When the climate got warmer, our ancestors migrated to mid-latitude Eurasia, where food availability was more seasonal than in Africa. These more dispersed food resources required increasingly sophisticated navigation,

stones to use as tools (Hublin et al., 2017). At about the same time, Middle Stone Age blades and spears with higher precision and sophistication were manufactured by protohumans in today’s Kenya. These were more abstract than earlier hand axes and meat scrapers that still preserved the shape of the original lump of rock. The leftover stone dust from these new tools was used paint the body of the creator, as a symbol of individuality or group identity (Brooks et al., 2018).

14. In fact, we do not have to go back in time literally. There are still a few uncontacted peoples left on our planet who live like their ancestors did 50,000 years ago. The North Sentinelese people who live in the Andaman archipelago between India and Malaysia do not know how to make fire and still use spears and arrows to hunt.

15. Boyd et al. (2011).

16. From flaking stones to mastering sharp bifacial hand axes requires multiple steps and pre-meditation. Tools exhibit a mental template imposed on unorganized, dead material. The large numbers of artifacts found in the Olduvai Gorge, West Africa, show a continuum of refinement (Toth, 1985). Capuchin monkeys use a rock and a log as a hammer and anvil to crack palm nuts for food in the cerrado of Brazil. Some monkeys bang rocks together and break off stone flakes. Big brains are thus not required for intentionality (Proffitt et al., 2016).

memory, and planning.¹⁷ The hunter-gatherers slowly transformed into farmers, established permanent domestic spaces, and invented more elaborate artifacts, including weapons, paintings, sculptures, and musical instruments. Many of these artifacts no longer mimicked or represented anything real in the world. They had to be imagined before their maker could start working. Artifacts are externalized versions of a thought, a reflection of contemplation, and a way to communicate personal knowledge to others even after the creator has vanished. Artifacts are semantic entities, which can be labeled and remembered as separate from other things. The root of this mirroring between action and perception may be built upon mechanisms analogous to active sensing, with its corollary circuits (Chapter 3).¹⁸ However, the feedback in this extended loop is not a specialized circuit within the brain but a sequence between action-produced artifacts and their reflection back to the brain.¹⁹

These externalized mental products come to exist outside their creator's brain as permanent social memory, enriching the collective mind. The creative action of tool-making might be considered a prelude to verbal language. Conceiving a multiple-component tool, such as a rock tied to a rod to make a hammer, requires a hierarchical algorithm, a natural precursor for language syntax. The manual dexterity required for tool-making is also necessary for subsequent writing ability.²⁰ Artifacts and, eventually, words can readily communicate semantic information from one brain to many without laborious episodic explorations by each individual (Chapter 5). Instead, the grounding of meaning is simply achieved by guidance or approval from others. Externalized information can be named, and, therefore, it rapidly spreads semantic knowledge. This ability comes with a cost, though. We accept the definition of events and phenomena too often without personal experience, accumulating and

17. Some view that such ecological factors comprised a critical evolutionary pressure and selection on human brain size (e.g., Parker and Gibson, 1977; Gonz  les-Forero and Gardner, 2018).

18. A good part of archeology is "neuroarcheology," a fossil record of thought (Malafouris, 2009; Renfrew et al., 2009; Hoffercker, 2011). Human-created objects not only document the migration of early people but also are a rich source of the intentions, lifestyles, and cognitive abilities of their masters. However, tools carry meaning only to those who share the "cipher" and can envision their use.

19. This loop is analogous to the way we do experiments in the lab. A thought is acted out by producing an experiment (e.g., making a lesion in the brain) and then watching and interpreting the consequences of our perturbation.

20. Greenfield (1991) hypothesized that the same neuronal substrate can serve the syntax necessary for both multiple-step object-making and construction of sentences. Although there is an analog of Broca's area in great apes, it can only support the complexity of both tool construction and symbol manipulation at the level of a two-year-old human (Holloway, 1969).

using a huge vocabulary in which we do not understand the true meaning of many words (Chapter 1).

Urban Revolution

Once hunter-gatherers started farming and domesticating plants and animals about 12,000 years ago, group knowledge began to increase dramatically. Agriculture required division of labor, specialization, hierarchy, and trade, associated with an acceleration of demographic growth. Tens of thousands of people lived together in the Sumerian cities of Uruk and Shurupak 6,000 years ago. The pressure on dwelling solutions led to the introduction of geometric designs that were entirely the product of the externalized mind, such rectangular and square house floors. As population density grew and cities were formed, the collective sharing of external thought expanded explosively. A city of 10,000 people working as a cooperating group can generate much more accumulated knowledge and much more rapidly than a hundred small communities of 100 people each.²¹ This stage of *H. sapiens* history, known as the *Urban Revolution*, brought us writing and mathematics.²² Kings, pharaohs, and their bureaucrats emerged, and the texts they created gained authority. The earliest pictographic or ideographic scripts gradually transformed into Sumerian wedge or cuneiform writing over a thousand years and then to the current forms of writing. Pictographs worked mainly as a mnemonic device to help with spoken language and had to be interpreted rather than read. They are a bit like the private notes and symbols I scribble in the margins of the books I read. Even I have a hard time interpreting them properly without consulting the related text. Yet pictographs are very useful for storing thoughts outside the brain. Such externalized thoughts, collectively agreed upon by a small group of powerful people, became the rule of the land for generations. Written texts served to remind members of the society that they must obey certain laws and that breaking them would result in punishment. The rulers invented money

21. Not only humans but the cognitive abilities of other animals also benefit from the size of a social group, including non-human primates (Humphrey, 1976) and Australian magpies (Ashton et al., 2018).

22. Because of the relatively simultaneous emergence of writing and math in Mesopotamia, several scholars have speculated that they required the same kind of cognitive processes and perhaps brain regions. However, this cannot be fully correct. The Inca people in Mesoamerica built a civilization comparable to that of Egypt, built pyramids with the help of sophisticated math and geometry, and had a method of record keeping with knotted cords but did not invent writing (Ascher and Ascher, 1981).

that allowed exchanges across variable services and products of labor. This externalized metric of value enabled further stratification of work and allowed even more people to cooperate and work together.

Cognitive Revolution

The ability to read the unique minds of other individuals through externalized objects, symbols, texts, and institutions gave rise to the cognitive revolution phase of human evolution. This sort of information reflects back to the brain to profoundly affect its circuits. In a modern experiment, illiterate women in India learned how to read and write their mother tongue, Hindi, while their brains were repeatedly imaged. After just 6 months of literacy training, several areas of their brains reconfigured, including the thalamus and even the brainstem.²³

Fast forward to Johannes Gutenberg's invention, which enabled the production of printed books in the fifteenth century²⁴, and then jump to the rotary presses of the nineteenth century, which introduced printing on an industrial scale. Written sentences, mathematical equations, and pictures are externalized products of the collective cognitive powers of the human mind, and they have an unlimited lifetime. Books allow people to check and compare the thoughts of others whose ideas were developed under different circumstances. Unlike a busy college professor, a book can provide all the time that the reader needs. The reader can consult its contents over and over again. Only the possession of the cipher is needed to convert the abstract patterns of letters and words into organized neuronal activity to create novel thoughts in the brain. Externalization of brain function enables both broadcasting and copying of complex ideas, intentions, emotions, and hopes from brain to brain without the need for immortality of the ideas' creator. By virtue of the hierarchical, generative, and

23. Previous studies focused on changes mainly in the early visual system (Carreiras et al., 2009; Dehaene et al., 2010). In the study of Skeide et al. (2017), the effects of literacy learning also extended to V3 and V4, the thalamic pulvinar, and the superior colliculus of the brainstem. The coupling of blood oxygen level-dependent (BOLD) signals between these subcortical structures and visual areas V1 to V4 increased after the acquisition of basic literacy skills. The Devanagari script of Hindi is an alpha-syllabic writing system that represents sound simultaneously at the syllable and phoneme level. The involvement of the superior colliculus might be a consequence of the fine-tuning of the initiation of saccadic eye movements necessary for fixating the eyes to relevant parts of the printed text (Dorris et al., 1997). The importance of the action system is also illustrated by the finding that visuospatial motor skills predict reading outcome in preliterate children (Carroll et al., 2016).

24. It is hard to be first in anything. Books had been printed in China and Korea for centuries before Gutenberg.

recursive nature of language syntax and its virtually limitless scope of expression, people can find their way to God or forbidden thoughts, come up with new hypotheses, and publish earth-shaking discoveries.

EXTERNALIZATION AND INTERNALIZATION ARE COMPLEMENTARY PROCESSES

As discussed in Chapter 5, disengagement of the brain from its environment and its ability to manipulate models of the external world are prerequisites for cognition. This operation is facilitated by externalization: internally generated thoughts are tested against action-produced artifacts, which reflect back on the original thought as a validating, grounding process. This is a protracted action–perception mirroring loop. We test the validity of a thought by acting in certain ways and, as a result, seeing things differently. Externalization is an elaborate test of internal thoughts. With the help of artifacts, such as books, thoughts can be communicated to many other brains. This practice can give rise to agreed abstraction or semantic knowledge within the community. Human-made objects are therefore a medium through which semantic knowledge and abstract ideas can rapidly spread.²⁵ No thought or concept possesses validity unless it is communicated to other brains. Meaning is created when a group of people weaves together a common story: “my thoughts are not so crazy.” Thus, a concept becomes a concept only by externalized grounding, and the process of grounding involves interactions with other brains. This extended view of cognition implies that cognition can be understood only as a social phenomenon that transcends the brain of an individual.²⁶

Inventing Numbers

How is a communicable abstract idea created? Nobody knows for sure how it happened first, but we can make up a plausible scenario. Let’s suppose that a smart individual wanted to throw a party for the village. As this person sacrificed one sheep after another, he scratched a line on a bone for each sheep. Tally marks were used for counting before the construction of cities in Mesopotamia.

25. The meaning of the Latin verb *abstracte* is to “remove, detach, or pull away.” A cave painting or other nonpractical artifact fuses the essential features of multiple concrete situations by detaching the conditions in which the concrete events occurred (Chapter 5).

26. The term “con-sciousness” literally means shared knowledge and is meant in this way in many languages.

After this process happened many times, the idea may have struck him that five vertical tally marks resembled five fingers. With such an insight, the next time no groove marks were needed; he just counted each sheep with a finger. He or his descendants who learned this approach might have called five sheep a “handful.” At this stage, there was still a match between symbols and the physical things they represented. Yet this was already a major abstraction that likely took several hundred years.

From here, archeology can assist us in reconstructing the evolution of externalized thought. For hundreds of years, arbitrary number systems were used to count different groups of things, such as animals and bags of grains, while other systems were applied to count other things.²⁷ People in different cities also used different tally mark systems. The next level of abstraction was the realization that numbers apply equally to all things. Once this generalization was made, tally marks could be manipulated without knowing what they referred to. Through such mental manipulation, the marks were separated from the circumstances that created them, and they became explicit knowledge. Numbers thus lost their dependence on their physical correspondences. Recall that this mental transformation is similar to the process of making maps from exploration or creating semantic knowledge from multiple personal episodes (Chapter 5). Thus, externalizing ideas in the form of tools and objects could have helped to generate novel thoughts, which, in turn, could be internalized into other minds. The initial process may have taken a thousand years, but it was accelerated rapidly by the need for many individuals to do the same activities again and again in increasingly large cities. No genetic change in brain hardware was needed to gain this ability.

We can witness the evolution of abstraction at an accelerated pace when observing our children. First-graders can count five apples, five pencils, and five other things by touching and holding them. Then they learn that “5” corresponds to the word “five” but often ask: “Five what?” as if looking for the grounding role of the objects. Adding concrete objects is a relatively easy task when kids can manipulate them. In the internalization phase, when counting imaginary apples, they touch their fingers prior to gradually treating “5” as an object and eventually as an abstract number. The entire process is speeded up not (only) because of brain maturation but because learning numbers is supervised by experienced adults whose brains have already abstracted the concept of numbers.

27. Clegg (2016).

Many mathematicians and physicist believe that numbers and math are the fundamental organizers of the universe, so that we could simply infer them from our mental imagination, independent of the outside world.²⁸ In this chapter, I have tried to make the case that the process acts in the opposite direction. Without externalization and human-made artifacts, it is unlikely that the concept of numbers and arithmetic would ever have emerged.

CLOCKS FOR SYNCHRONIZING HUMAN COOPERATION

Perhaps no externalized artifact has had a larger impact on our ideas than the clock. Vague concepts of space and time predated measuring instruments, mainly to explain the vastness of the surrounding world and our place in it.

Seasonal changes have predictable effects on the accessibility of various food resources and on agriculture. In addition, daily circadian changes influence the activities of all living creatures. Early humans routinely used celestial “time keepers,” such as the relative position of the sun and the moon, but activities can be better allocated within these natural time windows if shorter, discrete intervals can be determined. From ancient Egypt came the idea that the day from sunrise to sunset and the night from sunset to sunrise should be divided into twelve hours. There are different explanations for the origin of the number twelve. There are roughly twelve moon cycles in a year, each cycle corresponding to one complete orbit of the earth around the sun, and twelve zodiacal signs. Also, there are three phalanges in each of the four fingers. When numbers emerged by making correspondences between things in the world and references in our body, marking the phalanges was likely a good method for counting twelve with one hand. The Babylonians used a duodecimal (twelve) and a sexagesimal (sixty) system. Why they chose sixty is not clear except that it is a nicely “divisible” number (by 1, 2, 3, 4, 5, 6, 10, 12, 15, 20). Furthermore, ancient astronomers believed that the Heavens comprised sixty concentric translucent spheres, the Earth being at the center. In geometry,

28. According to Roger Penrose (2004), numbers can “seemingly be conjectured up, and certainly assessed by, the mere exercise of our mental imaginations, without any reference to the details of the nature of the physical universe.” Max Tegmark (2014) argues “that our physical world not only is described by mathematics, but that it is mathematics. . . . Everything in our world is purely mathematical—including you.” But there is no alternative method to quantify the world out there. Thus, math as a tool cannot be falsified. Mathematics starts out with *axioms*, fundamental assumptions without a need of proof. Math often makes predictions that the real world cannot absorb.

a sexagesimal system works better than decimal numbers. Modern clocks are still based on the 12/60 method.²⁹

Time is intimately related to movement and geometry. However, dividing the interval between sunrise and sunset to calculate time is problematic because its length varies with seasons and locations. Noon time and the length of the hour, measured in the old way, were different in different regions of the vast Roman empire, where sundials were ubiquitous. The ancient Greeks realized this and defined the hour based on equinox days, when day and night are of equal length. This celestial duration served to calibrate the shadow-casting sundial. Yet this approach did not solve the time compression/expansion issue at different longitudes.

Concrete Time and Duration

Water clocks were invented at about the same time as sundials but for a different purpose. They served to measure the duration of certain sacred or bureaucratic activities as opposed to the time of the day. In Greece, water clocks (called *klepsydra*) were used in courts to limit the time allowed for presenting complaints.³⁰ They served as a calibration device to settle the dispute over whether days and nights are of different lengths in different seasons. Much later, when the geometrical relationship between the height of the water in the tank and volume was understood, water clocks were used to calibrate the hour (*horos*).

In medieval Europe, churches were built with a bell tower to solicit the faithful to attend Mass.³¹ The monasteries became the first institutions to use mechanical clocks, which helped to organize their complex bureaucratic

29. Dohrn-van Rossum (1996). Of course, water clocks do not measure time flow per se but the volume of water drained from the container, which is then converted to time units calibrated by other measures.

30. Lamps fueled by oil were used in magical rituals. The volume of oil placed in the lamp determined the duration of the event. This ritual still exists in churches where the duration of the offered prayer is timed by a burning candle. Lamps, water clocks, and similar instruments measured duration only and were only much later synchronized to some celestial event to tell the time of day. Demonsthenes demanded “Stop the water!” when his court proceedings were interrupted by depositions (Hannah, 2009a, 2009b). In fact, the word “water” (*hidor*) was synonymous with time. Of course, water clocks do not measure time flow per se but the volume of water drained from the container, which is then converted to time units calibrated by other measures. Thus, the measurable volume grounds the abstract duration.

31. In 1456, the Hungarian-Serbian Army, led by János Hunyadi, defeated the Ottoman Turks besieging Belgrade (then Nándorfehérvár). The victory was important because it prevented the expansion of the Ottoman Empire for seventy years. In recognizing the importance of the victory, Pope Callixtus III ordered the bells of all churches to be rung every day at noon as a call to

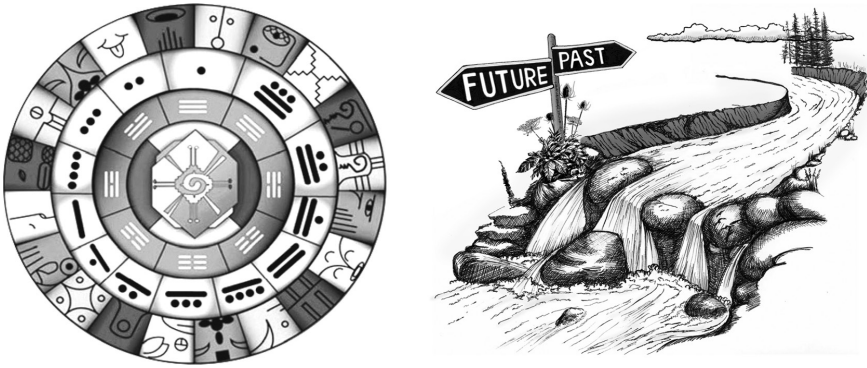


Figure 9.1. Conceptualization of change or lack of it. *Left:* Maya calendar. Life is a circle, nothing ever changes, everything repeats periodically. This view is shared by Hinduism, Buddhism, and Egyptian cultures. *Right:* Time flows, nothing is ever the same (*panta rhei*), a view adopted by Aztec, Christian, Jewish, and Islamic religions. Upstream is past; downstream is future. Relative position determines time flow.

apparatus. In the Muslim world, *müezzins* recited the call to prayer from the minarets at approximately the same local times of each day. Mainly as a result of such practices, the hour acquired a practical meaning both as duration and time of the day. In turn, it took centuries and numerous steps of innovation³² before mechanical clocks became accurate enough to divide hours into minutes and later into seconds. Today's atomic clocks are based on the duration of a particular number of energy transitions of the cesium atom. However, the discrepancy between instrument-kept time and time based on astrological movements persists even today. Leap seconds need to be inserted every now and then to compensate for the mismatch.

Internalization of Externalized Time

The increasing prevalence and precision of measuring instruments changed the human concept of time. Egyptian and other early societies did not see time as an arrow. Instead, change was envisioned as cosmic cycles (Figure 9.1). The wheeling of the sun, moon, and stars through the heavens meant time. Although they

pray for the souls of the defenders of the Belgrade battle. Of course, noon meant different times in different countries in old Europe. The noon bell ritual persists to this day.

32. The story of John Harrison, the English clockmaker who invented a marine chronometer and solved the problem of calculating longitude at sea for dead-reckoning navigation, has been told many times. My favorite account is by Dava Sobel (1995).

surely realized that humans and animals were born and died, this applied only to individuals. Species were eternal. This view was in harmony with the dominant ideology of preservation of the existing system—for almost 3,000 years in Egypt. Instead of historical explanations, powerful societies created a myth whose content equally applied to the past, present, and future. Past events, such as creation, were imagined as the actions of gods, but the distinction between past and future was inconsequential. In ancient Greece, Plato insisted that nothing is new, the soul already knows the truth, and everything is just reminiscence (*anamnesis*). “Nothing is new under the sun” (Ecclesiastes 1:9).

The concept of linearly progressive time is rooted in the Judeo-Christian view and the encouragement of manual labor and practical innovation, especially after the thirteenth century, as agricultural productivity and population density were rising in Europe.³³ The technology that was key to the emerging Industrial Age was the mechanical clock. The church embraced the clock, which allowed timing of prayers. By installing bells in virtually every community, these human-made devices created an “arbitrary temporal matrix within which everyone functioned. They generated a component of the daily landscape of the mind.”³⁴ The clocks and other instruments with their arbitrary units became metaphors for the universe and contributed to the mechanistic worldview of the Middle Ages. This new interpretation of the world was radically different from, say, the world of hunter-gatherer societies, which was filled with timeless ghosts and deities.³⁵ The linear concept of time gradually but fundamentally transformed the European mind, although it took centuries. The habit of thinking within the framework of time’s arrow for interpreting history is a relatively recent phenomenon.³⁶

Let us pause here for a second and summarize that the concept and meaning of time evolved considerably over the history of humankind. The externalization

33. Those who believed otherwise were punished. The atheist Giulio Cesare Vanini wrote in 1616: “human history repeats itself. Nothing exists today that did not exist long ago; what has been, shall be.” He was executed for these views by the church (as quoted in Wootton, 2015). Yet Christianity retained elements of recurrence. The liturgy is organized around the endless cycle of Christ’s life.

34. Hoffeecker (2011).

35. This connection was first articulated most clearly by Gordon Childe (1956). [NeuroNote: Childe suffered from social anxiety.] While rods were used routinely for measuring distances while building pyramids and water clocks were used to divide the day into hours by the aristocrats of Egyptian society, it was not until the Middle Ages that everyday people began to have access to them. There existed a fully articulated philosophy of the universe, based on purely mechanical lines, which served as a starting point for all physical sciences (Mumford, 1934).

36. Collinwood (1946); Lévi-Strauss (1963); Hannah (2009a).

of brain function facilitated the emergence of concepts and the spread of abstract ideas. The more people interacted with measuring instruments, the more real their abstraction-based units appeared to be. By measuring distance and duration, we formed a more concrete concept of space and time.³⁷ That is to say, emerging societal knowledge can shape concept formation in individuals. In support of this hypothesis, children below the age of nine intuitively grasp the notion of speed but not time. When one toy train is moved faster than another but for the same duration, children most often respond that the faster train traveled for longer time. The idea of velocity is natural at a young age, whereas the concepts of space and time are acquired from adults.³⁸

Humankind has spent considerable effort in perfecting time-keeping instruments. The clock has become the ultimate externalized synchronizer of human thought, with a tremendous impact on our concepts of the world and also on our activities (see Chapter 10). Without precise clocks, we could not conceive of travel, communication, and cooperation at today's global scale.

Paradoxically, the idea of clocks as servants of humans changed radically starting with the Industrial Revolution. Clocks became tools to measure productivity and regulate labor, and through this process time got linked to value. The abstract concept of time has become concrete money. However, the worker's sense of time and the time marked by the manager's clocks always compete forcefully. The tyranny of clocks is something we do not fully understand. Clock- and calendar-mandated demands, from morning alarms to grant deadlines, place extraordinary pressure on our daily lives. Over the past century, time has been perhaps the most important factor in advancing technology while also being the number one burden on human endeavors. This is strange because we know that rats, monkeys, humans, and likely other species perform worse under time pressure than without it.

THE INFORMATION REVOLUTION PHASE

Externalized thought in the form of objects, books, and communication media allows the spread of an individual's knowledge to many people. The process is

37. "By the use of a clock the time concept becomes objective" (Einstein and Infeld, 1938/1966).

38. "Time appears as a coordination of velocities" (Piaget, 1957). Jean Piaget concluded that temporal concepts are understood much later in development than are spatial concepts, whereas the idea of velocity is present from early on: "time relationships constructed by young children are so largely based on what they hear from adults and not on their own experiences" (Piaget, 1946).

reciprocal because the knowledge of many brains can also be communicated to a single individual. Books, computers, and Internet communication externalize brain functions at a grand scale and provide virtually unlimited storage space for accumulated human knowledge. However, like books in a library, this knowledge is only as useful as its accessibility. When I was a student, I looked up references related to my work and ordered the articles through interlibrary loan. When the book or journal arrived a few months later, I had to request permission in writing to make a print copy for myself. This process is much easier today, but our self-imposed demands have increased proportionally since rapid access is available to everyone.

In 1992, it took me forever to filter traces and compute the Fourier transform of a single channel electroencephalogram (EEG) on my Macintosh Classic II with its 80 Mb hard drive and 10 Mb of RAM. Today, Google's DeepMind can teach itself how to beat humans at Go and Atari games.³⁹ The FaceNet system can recognize a particular face among a million others. Similar systems can locate street scenes anywhere in the world and do a better job than groups of well-traveled humans. IBM's Watson and its brothers are trained to outperform doctors with decades of experience in diagnosing diseases.⁴⁰ By 2020, 80% of the adult population worldwide will own a smartphone that provides access to nearly everything humankind knows. Self-driving vehicles are around the corner. Wearable sensors monitor the activities of more than 50 million registered users, and new classes emerge daily. Monitoring the pattern of strikes on the keyboard or tracking eye movements, pupil size, and heart rate during reading of tens of millions of users provides an unprecedented amount of analyzable and quantifiable data. While you are being interviewed for a job or reading this page, sensors can collect information about your reactions, health states, and even desires, generating unprecedented amounts of biometric data that will be eventually owned by unknown groups of people. Artificial intelligence (AI) technologies will transform nearly every aspect of life from agriculture and medicine to transportation and finance. Tomorrow's AI machines will learn on their own, rivaling human performance in surprising and unforeseen domains and complementing the human brain.

39. See Silver et al. (2016). An excellent review about deep learning and neuroscience-inspired artificial intelligence is by Hassabis et al. (2017). A longer treatment of the topic is by Sejnowski (2018).

40. The cash flow of Apple, Microsoft, and Facebook exceeds by tenfold the annual federal budget for biomedical research in the United States (Insel, 2017). These rapid changes bring about a massive transformation of the workforce. Today, only 2% of Americans work in farms and 20% in industry, while the remaining part of the population provides services, with an increasing share of those working with "information" (Harari, 2017).

The Internet is becoming a world communication language, the new *agora* for exchanging externalized knowledge at a global scale. Words, expressions, and idioms can already be translated into virtually any language instantaneously, and the translation programs are getting better every day. We can internalize new knowledge at an unprecedented speed, test “what will happen if” scenarios without leaving our chairs, and, in the process, make new discoveries. Yet such globalization comes with homogenization. We are guided to follow the same trends and brands, whether we live in the United States or the Philippines. For a small, short-term benefit, we reveal our personal lives voluntarily on the Internet, encouraged by the many companies that benefit from learning about our needs, goals, and income.

Neurotechnology: The Next Revolution

As a culmination of the externalization of brain function, science has become perhaps the most powerful driving force of societal change. Perhaps the next new era in human evolution will be neurotechnology, the development of noninvasive techniques for monitoring and manipulating brain circuits with high spatial and temporal precision. Getting information about neuronal activity indirectly from the brain at relatively slow speeds is already possible with functional imaging. High-density recording of electrical and magnetic signals generated by large aggregates of superficial cortical neurons at the speed of neuronal communication is also routine. There are still a number of hurdles blocking the next step, recording from representatively large numbers of neurons with single action potential resolution and manipulating critical numbers of neurons one by one or in small groups. But, once implemented, such externalizing brain activity at high speed will allow the design of feedback actuator mechanisms to reshape the firing patterns of neuronal assemblies brought about by disease or substitute for the actions of missing neurons in brain repair.

Signals representing anything from brain states to the intention of one brain can be routed to many other brains, bypassing the current “slow” brain-to-brain communication via language. In speech, only one speaker speaks at a time, and others respond one by one. In direct brain-to-brain communication, we could download thoughts, intentions, and emotions from several brains and route them to many other brains, allowing for collective thinking and consensus manipulation. Messages sent and received on Twitter and Instagram are already precursors of semi-online multibrain communication. New senses for magnetic fields, infrared and ultraviolet light, or radioactivity, could be incorporated into the brain. While such “telepathic” communication sounds futuristic at the

moment,⁴¹ the laws of physics do not warn us that such technology is impossible. Rather, the big question is who needs such technology and what benefits it offers. The development of these techniques is likely to be different depending on whether they are driven by scientific curiosity, therapeutic considerations, or commercial interests.⁴²

BRAIN RESET

When the Ancient Library of Alexandria burned down, an enormous amount of hard-earned knowledge vanished forever. Imagine waking up in the morning to learn that all knowledge that existed in digital form has disappeared. That loss would reduce humankind's externalized knowledge to the level of the pre-Sputnik era. Almost no one could build a computer or design a search engine from scratch. Today's technologies require the cooperative efforts of many highly trained individuals, all of whom gain their smartness from the fast accessibility of externalized human brain products.

So who is smarter? An average hunter-gatherer who depends on a wide variety of skills to survive in the African savannah, or an average New Yorker who has a cell phone, can call a cab, finds a Starbucks at every corner, and orders takeout food after spending eight hours in an air-conditioned workplace? Our easy access to the vast accumulated knowledge of humankind creates the illusion that our brain's performance increased over the past 50,000 years. Yet our stand-alone brain is just the same hardware filled with different kinds of knowledge, updated to current needs. When we search for the superiority of the brain of *H. sapiens* over other animals, we need to understand not only neuron number, connection density, and clustering rules, but also how human brains externalize information to learn from the experience of others. Externalized knowledge of a community of minds allows for efficient internalization, a prerequisite for redesigning ourselves.

There is a price to pay for these gains. While the knowledge of humankind increases exponentially due to the multiplicative interactions of many brains, the relative share of the individual decreases dramatically, leading to an

41. Entertaining accounts of such futuristic perspectives are by Harari (2015) and Tegmark (2017).

42. The driving force of new technologies has recently shifted from scientific curiosity to commercial interest. Especially in the case of applied and translational science, the decision-making committees are increasingly composed of managers and business executives who tend to support proposals for technologies that rich customers like themselves can buy (Dyson, 1997).

increasing division of labor. As a systems neuroscientist, I am not an expert in molecular issues, neuroimmunology, or artificial intelligence, and my knowledge of the numerous diseases of the nervous system is rudimentary. It is hard to be a polymath these days, even within the confines of a single discipline. This situation, however, does not mean we should not try. The knowledge is there, waiting to be absorbed.

Perhaps some of my thinking is misguided. We find ourselves simultaneously saddened and enthusiastic when we learn about a new robot that outperforms a human in balancing skills, memory, or other cognitive games. However, this is not a fair comparison. The robot or supercomputer was produced by hundreds to thousands of interactive human brains. No wonder it can beat a single *Homo sapiens*. We often mix up the comparison between human versus machine knowledge and the knowledge of humankind versus “machinekind.” But the argument can go the other way as well. Our knowledge is rich not only because of the performance of the human brain but also because there are 7.4 billions brains on this planet, and a good fraction of them can communicate effectively, facilitated by the Internet. It does not seem realistic to believe that energy-hungry robots can build billions of other robots, each comparable with the human brain.

This is not to say that AI will not undergo an unprecedented and breathtakingly spectacular revolution. There is virtually no remaining frontier of brain-based intelligence that has not been challenged by AI in one way or another. Just a decade ago, it seemed unrealistic that face or scene recognition could be achieved by a machine. Today, they far surpass the best human experts. Their “reasoning” ability is demonstrated by beating every human in chess and other games.

Imagination and creativity? Hm. What are these faculties? Writing this book required reading tons of works by others, extracting and combining existing information, and creating a novel story by my brain. Can AI accomplish this? Nothing says it cannot. AI can already do such things by writing interesting stories, transcribing text to speech and vice versa, translating from one language to another better than most of us, composing new music that many of us can enjoy, and creating esthetically pleasing paintings. Many of these accomplishments have been suggested to be inspired by the brain. Yet this inspiration is not much more than analogy or a metaphor since the hardware realization of the machine algorithms are so different in brains and computers. If the hardware base is so different, then the algorithmic solutions are likely to be very different as well. And one more thing. The brain has both intelligence and emotions. Emotions do not rise in the brain alone but also require the rest of the body as well. There is no such thing (yet) as artificial emotion (AE), even if it is fun to imagine otherwise. For this simple reason, the machine and algorithms

do not have a first-person perspective on anything. A robot has no agenda. Current AI devices are autistic in the sense that they cannot read intentions or deduce our beliefs from our actions. Even if future-generation machines can read our minds and respond accordingly, they do not take such messages “personally.” They remain obedient partners, like well-trained seeing-eye dogs.

Therefore, I am not scared that self-organized AI machines will turn against us, as is often depicted in science fiction movies.⁴³ For robots, happiness, loneliness, and pain are absurd concepts. They have no desire to clone themselves, work together with other robots and with us, or develop agendas against humans. In the unlikely case event that they do, we can just pull the plug, and they will become lifeless and useless objects. Machines are not dangerous. We humans are.

A key aspect of externalizing brain function is the creation of measuring instruments and conceptualization of their abstract units. These instruments, such as clocks, accelerated the formation and communication of novel concepts and affected how we view the world and ourselves. Two concepts appear fundamental to our current world view: space and time, whose relationship to brain activity we discuss next.

SUMMARY

The brain areas in charge of generating plans and thoughts share many similarities with the motor cortex in terms of cellular architecture and input–output connectivity. The main difference is that prefrontal cortex does not directly innervate motor circuits. Instead prefrontal cortical areas can be designated collectively as an internalized action system, and thus plans and thoughts can be conceived of as internalized neuronal patterns that serve as a buffer for delayed overt action. Thoughts are only useful if they lead to action, even if that action is delayed by days or years. These same brain areas and mechanisms are also responsible for externalizing thought in the form of artifacts, language, art, and literature. In turn, externalized objects, as the tangible products of abstract thought, can have a profound impact on the creator’s mind and on the minds of others. Thus, externalized brain function facilitates the communication of explicit knowledge, hard-earned by a few, to all members of the community, enabling the quick and efficient spread of semantic knowledge.

43. Futurist forecasters always make the mistake of including timelines in their prophecies. Arthur C. Clark (1987) predicted thirty years ago that, in 2017, we would ride orbiting space stations and reach old age with the strength and energy of a teenager, but he failed to predict the Internet or the collapse of the Soviet Union.

This effective externalization of brain function—not special wiring or other qualities of our nervous system—explains why we became the rulers—and potentially the destroyers—of our planet. The cognitive evolution of the human race started with tool-making and language and transitioned through an Urban Revolution to our present Information Age. In the process, the individual's relative share of humankind's collective knowledge has decreased dramatically, driving the need for a increasing cooperation for survival. Without extensive cooperation and externalized brain products, our knowledge would be set back to the level of our hunter-gatherer ancestors.